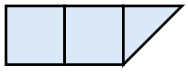
# Preview: set-01/by_level/Grade_03-04_Ecolier/y2023_q09 (v1)

Filling a shape with pieces: start where there is least room (corner, narrow arm), match slanted edges, check with the number of squares

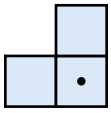
***Warm-up: small steps first (no points)***

**W1.** How many half squares (triangles made by cutting a square in two) make up this piece?



***A.*** 3 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

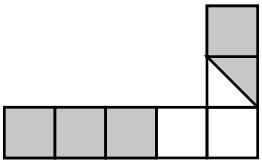
**W2.** Two pieces fill this shape. Which piece lies on the dot? The pieces may be turned.

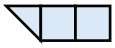


***A.*** ***B.*** ***C.*** ***D.*** ***E.***

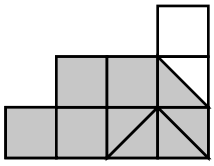
***Part A: Each question is worth 3 points***

**1.** Some pieces are already in the shape (grey). Which piece fills the white gap exactly? The piece may be turned.



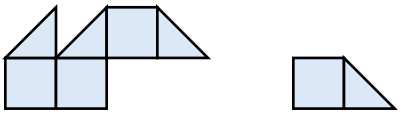
***A.*** ***B.*** ***C.***  ***D.*** ***E.***

**2.** Some pieces are already in the shape (grey). Which piece fills the white gap exactly? The piece may be turned.



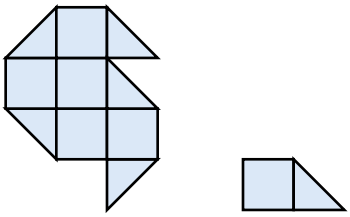
***A.*** ***B.*** ***C.*** ***D.*** ***E.***

**3.** How many pieces like the small one are needed to fill the big shape completely? The pieces may be turned.



***A.*** 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 9

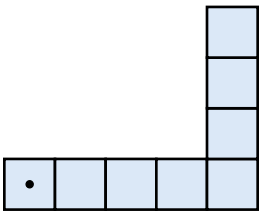
**4.** How many pieces like the small one are needed to fill the big shape completely? The pieces may be turned.



***A.*** 3 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 15

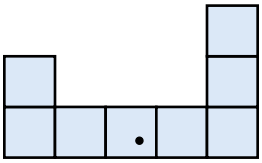
***Part B: Each question is worth 4 points***

**5.** Using the pieces A, B, C, D and E you can fill this shape completely. The pieces may be turned. Which of the pieces lies on the dot?



***A.*** ***B.*** ***C.*** ***D.*** ***E.***

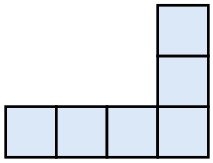
**6.** Using the pieces A, B, C, D and E you can fill this shape completely. The pieces may be turned. Which of the pieces lies on the dot?



***A.*** ***B.*** ***C.*** ***D.*** ***E.***

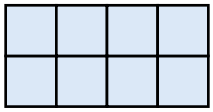
***Part C: Each question is worth 5 points***

**7.** Four of the pieces A, B, C, D and E fill this shape completely. The pieces may be turned. Which piece is left over?



***A.*** ***B.*** ***C.*** ***D.*** ***E.***

**8.** Four of the pieces A, B, C, D and E fill this shape completely. The pieces may be turned. Which piece is left over?



***A.*** ***B.*** ***C.*** ***D.*** ***E.***

# Answer Key

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| W1. | **B** | **W2.** | **B** | **1.** | **D** | **2.** | **A** | **3.** | **C** | **4.** | **C** |
| 5. | **C** | **6.** | **C** | **7.** | **B** | **8.** | **C** | | | | |

### *Solutions*

**W1. B** (5) — A whole square is 2 half squares. Counting gives 5. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · w_count_halves]

**W2. B** — Start in the corner with the dot and try each piece. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · w_corner_first]

**1. D** — Count the half squares of the gap (5) and look at where its slanted edges are. Only piece D has the same shape. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · missing_piece]

**2. A** — Count the half squares of the gap (3) and look at where its slanted edges are. Only piece A has the same shape. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · missing_piece]

**3. C** (3) — Count half squares: the shape has 9, one piece has 3. 9 ÷ 3 = 3. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · how_many_tiles]

**4. C** (5) — Count half squares: the shape has 15, one piece has 3. 15 ÷ 3 = 5. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · how_many_tiles]

**5. C** — Start where there is the least room: the corners and the narrow parts of the shape. Slanted edges must meet slanted edges. When all pieces are in, piece C lies on the dot. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · dot_piece]

**6. C** — Start where there is the least room: the corners and the narrow parts of the shape. Slanted edges must meet slanted edges. When all pieces are in, piece C lies on the dot. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · dot_piece]

**7. B** — Pieces A and B have the same number of half squares, so counting alone does not decide. Fit the pieces in, starting in the corners: piece B is not needed. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · left_over]

**8. C** — Pieces E and C have the same number of half squares, so counting alone does not decide. Fit the pieces in, starting in the corners: piece C is not needed. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · left_over]